

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC Course

Extension 1 Mathematics

Assessment 2 March 2015

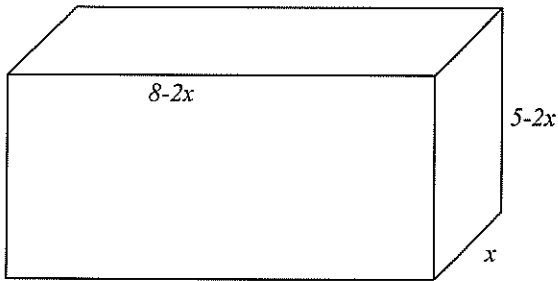
TIME ALLOWED: 70 minutes

Instructions:

- Write your name and class at the top of this page, and on your answer booklet
- Both this question sheet and the answer booklet must be handed in
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- Approved calculators may be used.

SECTION 1: MULTIPLE CHOICE (5 Marks)

Write your answers on the Multiple Choice Answer Sheet, included in your answer booklet.
All questions are worth 1 mark

1	Joan pays back her bank loan of \$105 000 in 10 years with equal monthly payments of \$1 500. Her equivalent simple interest charge would be: A. 4.17% p.a. B. 5.83% p.a. C. 7.14% p.a. D. 17.14% p.a.
2	The curve $y = x^4 - 2x^3 - 12x^2 + 12x - 2$ is concave up for: A. $-1 < x < 2$ B. $x < -1$ C. $x > 2$ D. $x < -1$ or $x > 2$
3	When the area between the curve $y = \sqrt{x(x^2 - 9)}$ and the x-axis is revolved about the x-axis, the volume of the solid formed is given by: A. $\pi \int_{-3}^0 x(x^2 - 9)dx$ B. $\pi \int_0^3 x^2(x^2 - 9)^2 dx$ C. $\pi \int_{-3}^3 x(x^2 - 9)dx$ D. $\pi \int_{-3}^3 x^2(x^2 - 9)^2 dx$
4	 A box measures x cm by $(5-2x)$ cm by $(8-2x)$ cm. The maximum volume of this box occurs when A. $x = 1$ B. $x = 10/3$ C. $x = 1$ or $x = 10/3$ D. $x = 2.5$
5	The value of $\int_{-4}^4 \sqrt{16 - x^2} dx$ is: A. 8π B. 16π C. $2\sqrt{2}\pi$ D. 0

SECTION 2

(START EACH QUESTION ON A NEW PAGE OF YOUR ANSWER BOOKLET)

QUESTION 6: (8 Marks)

- | | | |
|-----|---|---------------------------|
| | | Marks |
| (a) | Find indefinite integrals of: | 2 |
| | (i) $(x + 1/x)^2$ | (ii) $\frac{5}{\sqrt{x}}$ |
| (b) | Find the value of $\int_1^{32} \frac{4}{x^{1.4}} dx$ | 2 |
| (c) | Prove, by the method of Mathematical Induction, that | 4 |
| | $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2}{4}(n+1)^2 \text{ for all values of } n > 0$ | |

QUESTION 7: (8 Marks) Start a New Page

- | | | |
|-----|---|--------------|
| | | Marks |
| (a) | Differentiate $(1 + x^2)^3$ and hence find $\int x(1 + x^2)^2 dx$ | 2 |
| (b) | Using the Trapezoidal Rule with 3 function values, find an approximation, to 2 decimal places, for: | 3 |
| | $\int_1^5 \sqrt{25 - x^2} dx$ | |
| (c) | The area enclosed by the curves $y = x^3$ and $y^2 = 32x$ is rotated about the x -axis. | 3 |
| | What is the <u>exact</u> volume of the solid formed? | |

QUESTION 8: (8 Marks) Start a New Page

Marks

- (a) The power loss in a length of electrical wiring, in watts per km, is given by the formula

$$L = C^2 r + \frac{5}{r}$$

Where C is the current (in amps) and r is the resistance (in ohms)

- (i) For a given current, C , what is the resistance required to give a minimum loss of power per kilometre?

3

- (ii) What is the value of this loss? (in watts/km)

1

- (b) Jeffrey would like to save \$60 000 for a deposit on his first home. He decides to invest \$3 000 each month from his monthly salary into a bank account which earns interest of 6% per annum, compounded monthly. Jeffrey intends to withdraw \$ M from this account at the end of each month, straight after the interest has been paid, for living expenses.

- (i) Show that the amount in the account, following the withdrawal of the second set of living expenses is given by $\$6\,045.08 - 2.005M$

1

- (ii) Calculate, showing all working, the value of M , to the nearest dollar, if Jeffrey is to reach his goal after 5 years.

3

QUESTION 9: (8 Marks) Start a New Page

Marks

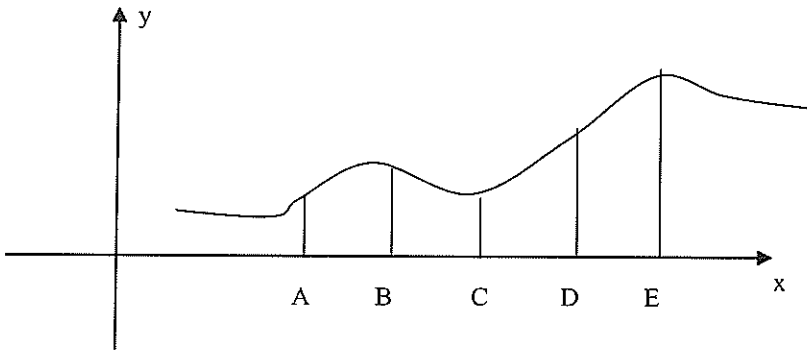
- (a) Using the substitution $u = 2x^2 - 1$, or otherwise, find the value of

4

$\int_1^3 \frac{x}{\sqrt{2x^2-1}} \, dx$ correct to 2 decimal places

- (b) A vase is formed by rotating the area between the curve $y = g(x)$, shown below, the x -axis, and the lines $x= 1$ and $x= 5$.

4



A table of values for the curve, at the points where $x = 1, 2, 3, 4$ and 5 is given below

Point	A	B	C	D	E
x	1	2	3	4	5
$g(x)$	3	5	4	6	8

Using Simpson’s Rule, with 5 function values, find the volume of the vase.
(Give your answer in terms of π)

QUESTION 10: (8 Marks) Start a New Page

Marks

For the curve $y = \sqrt{x}(4 - x)$,

(i) Give any restrictions on the Domain of x 1

(ii) Find $\frac{dy}{dx}$ 1

(iii) Find all turning point(s) and their nature. 3

(You may assume the result $\frac{d^2y}{dx^2} = \frac{-3x-4}{4x\sqrt{x}}$)

(iv) There are no inflexion points for this graph. *Explain why not.* 1

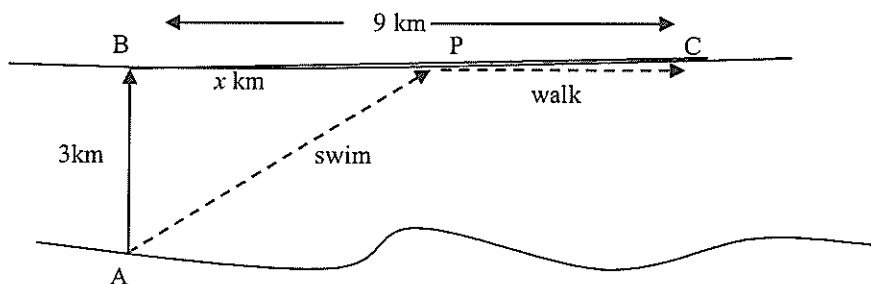
(v) Using all of the information above, sketch $y = \sqrt{x}(4 - x)$ 2

QUESTION 11: (8 Marks) Start a New Page**Marks**

- (a) B is a point across a river from a man at A, and is 3 km due North of A.

C is a position on the same side of the river as B, and 9 km due East of it.

The man at A intends to swim from A to a certain spot, P, on the opposite bank, where P is in a direct line from B to C, and x km from B.



- (i) He can swim at 4 kph. Show that the time taken for him to swim to P from A is

1

$$\frac{1}{4}\sqrt{x^2 + 9} \text{ hours.}$$

- (ii) The man walks at a speed of 5 kph from P to C.

1

Find the total time (T) for him to get from A to C, via P, by a combination of swimming and walking.

- (iii) Calculate the value of x which will minimize the time it takes him to get from A to C
(You must show all working)

3

(b) (i) Show that $1 - t + t^2 - \frac{t^3}{1+t} = \frac{1}{1+t}$

1

- (ii) Prove that, for $t > 0$ and $x > 0$,

2

$$\int_0^x \frac{dt}{1+t} < x - \frac{x^2}{2} + \frac{x^3}{3}$$

END OF THE EXAMINATION

MATHEMATICS EXTENSION 1

YEAR 12 2015

MULTIPLE CHOICE

1. C 2. D 3. A 4. A 5. A.

SECTION 2

(b) (a)(i) $\int (x^2 + \frac{1}{x} + 2) dx = \frac{1}{3}x^3 - \frac{1}{x} + 2x + k$

(ii) $\int \sqrt[3]{x} dx = \begin{cases} 10x^{\frac{1}{2}} + k \\ 10\sqrt{x} + k \end{cases}$

(b) $\int_1^{32} 4x^{-\frac{1}{10}} dx = -10x^{-\frac{1}{10}} \Big|_1^{32}$
 $= -10x^{-\frac{3}{5}} \Big|_1^{32}$
 $= -10\left(\frac{1}{4}\right) + 10(1)$
 $= 7\frac{1}{2}$

(c) For $n=1$, LHS = 1, RHS = 1
 $\therefore$ true for $n=1$

Assume the formula is true for $n=k$
is $1^3 + 2^3 + \dots + k^3 = \frac{k^2}{4}(k+1)^2$

For $n=k+1$

$$\begin{aligned} 1^3 + 2^3 + \dots + k^3 + (k+1)^3 &= \frac{k^2}{4}(k+1)^2 + (k+1)^3 \\ &= (k+1)^{\frac{1}{4}} [k^2 + 4(k+1)] \\ &= \frac{(k+1)^2}{4} (k+2)^2 \end{aligned}$$

which is of the same form.

$\therefore$ If the formula is true for $n=k$, it is true for $n=k+1$.

BUT it is true for $n=1$

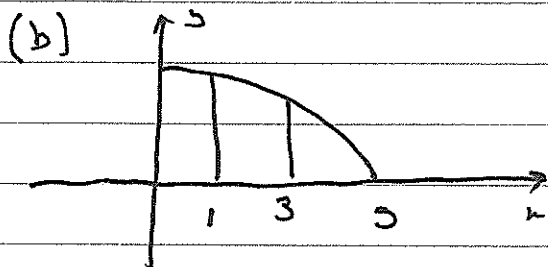
$\therefore$ $\dots \dots \dots n=2$ and so on

is true $\forall n$

7

$$(a) \quad \frac{d}{dx}(1+x^2)^3 = 6x(1+x^2)^2$$

$$\therefore \int x(1+x^2)^2 dx = \frac{1}{6}(1+x^2)^3 + k$$



$$T_1 = \frac{1}{2} \times 2 \times [\sqrt{24} + 4]$$

$$T_2 = \frac{1}{2} \times 2 \times [4 + 0]$$

$$\therefore A = \begin{cases} \sqrt{24} + 8 \\ 2\sqrt{6} + 8 \end{cases} \text{ OR } \begin{cases} 12.90 \\ 12.89 (6 \text{ sig figs}) \end{cases}$$

(c) The curves intersect at $x^6 = 32x$

$$\therefore x(x^5 - 32) = 0$$

$$\Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases} \text{ OR } \begin{cases} x = 2 \\ y = 8 \end{cases}$$

$$\therefore Vol_1 = \int_0^2 x^6 dx$$

$$= \left[\frac{1}{7} x^7 \right]_0^2$$

$$= \frac{128}{7}$$

$$Vol_2 = \int_0^2 32x dx$$

$$= \left[16x^2 \right]_0^2$$

$$= 64$$

$$\therefore Vol_{\text{req}} = \frac{448 - 128}{7}$$

$$= \begin{cases} 320/7 \text{ cu units} \\ 45 \frac{5}{7} \text{ cu units} \\ 45.71 \text{ cu units} \end{cases}$$



$$(8) (a) \quad \frac{dL}{dr} = c^2 - 5/r^2$$

$$(1) \quad \frac{d^2L}{dr^2} = 10/r^3$$

$$\text{At min} \quad \frac{dL}{dr} = 0$$

$$\therefore c^2 = 5/r^2$$

$$\therefore r = \sqrt{5}/c$$

$$\left\{ L'' > 0 \Rightarrow \text{min.} \right.$$

$\therefore$ min loss occurs when $r = \sqrt{5}/c$ ohms

$$(b) (i) \quad A_1 = 3000(1.005) - M$$

$$A_2 = [3000(1.005) - M]1.005 + 3000(1.005) - M$$

$$= 3000(1.005)^2 + 3000(1.005) - M(1.005 + 1)$$

$$= 6045.08 - 2.005M$$

$$(ii) \quad A_n = 3000(1.005)^n + 3000(1.005)^{n-1} + \dots - M(1 + 1.005 + \dots + 1.005^{n-1})$$

$$= 3000[1.005^n + 1.005^{n-1} + \dots + 1.005] - M[1 + 1.005 + \dots + 1.005^{n-1}]$$

$$\text{And } A_{60} = 60000$$

$$\therefore 60000 = 3000(1.005) [1.005^{59} + \dots + 1] - M [1 + 1.005 + \dots + 1.005^{59}]$$

$$M \left[\frac{1.005^{60} - 1}{0.005} \right] = 3000(1.005) \left[\frac{1.005^{60} - 1}{0.005} \right] - 60000$$

$$M = 3000(1.005) - \frac{60000}{\frac{1.005^{60} - 1}{0.005}}$$

$$= 3015$$

$$= 3015 - 859.97$$

$$\approx 2155.03$$

9

(a) $u = 2x^2 - 1$

x	1	3
u	1	17

$$\frac{du}{dx} = 4x$$

$$du = 4x dx$$

or) $\rightarrow du = \frac{du}{4x}$

$$\therefore \int_1^3 \frac{x}{\sqrt{2x^2-1}} dx = \int_1^{17} \frac{x}{\sqrt{u}} \cdot \frac{1}{4x} du$$

$$= \int_1^{17} \frac{du}{4\sqrt{u}}$$

$$= \left[\frac{1}{4} \cdot 2 u^{1/2} \right]_1^{17}$$

$$= \frac{1}{2} (\sqrt{17} - 1)$$

pl.)

$$\approx 1.56$$

(b)	A	B	C	D	E
x	1	2	3	4	5
$g(x)$	3	5	4	6	8
$g(x)^2$	9	25	16	36	64

$$VOL_1 = \frac{\pi}{3} \cdot 1 [9 + 16 + 100] \quad VOL_2 = \frac{\pi}{3} \cdot 1 [16 + 64 + 144]$$

$$VOL = \frac{\pi}{3} [349]$$

$$= \frac{349\pi}{3}$$

10

(i) $x \geq 0$

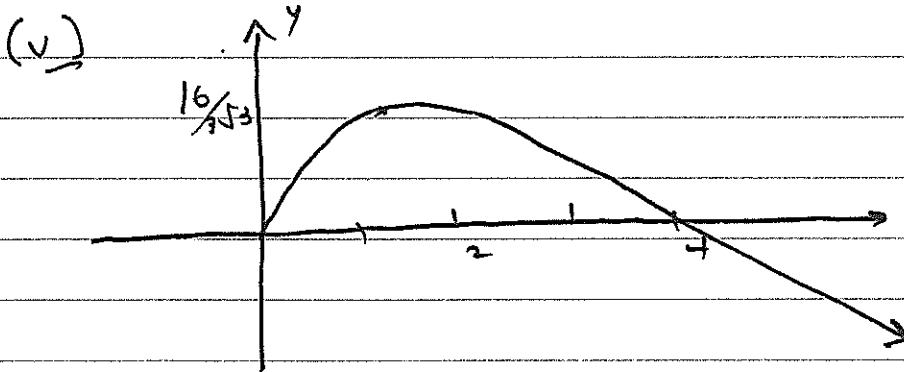
(ii) $\frac{dy}{dx} = 2x^{-1/2} - \frac{3}{2}x^{1/2}$
 $= \frac{1}{2}x^{-1/2} [4 - 3x]$
 $= \frac{4 - 3x}{2\sqrt{x}}$

(iii) At S.T. $\frac{dy}{dx} = 0$

$\therefore \begin{cases} x = \frac{4}{3} \\ y = \sqrt{\frac{4}{3}} \left(4 - \frac{4}{3}\right) \\ = \frac{16}{3\sqrt{3}} \end{cases}$

est $y'' < 0 \Rightarrow \text{max T.P. at } \left(\frac{4}{3}, \frac{16}{3\sqrt{3}}\right)$

(iv) Because for $y'' = 0$, $x = -\frac{4}{3}$ which is outside the Domain



(11) (a)(i) $AP = \sqrt{x^2 + 9}$

$$\text{Time} = \frac{\text{Distance}}{\text{speed}}$$

$$= \frac{\sqrt{x^2 + 9}}{4}$$

(ii) from P to C, time = $\frac{9-x}{5}$

$$\therefore \text{Time taken} = \frac{1}{4} \sqrt{x^2 + 9} + \frac{9-x}{5}$$

(iii) $\frac{dT}{dx} = \frac{1}{4} \cdot \frac{1}{2} (x^2 + 9)^{-1/2} \cdot 2x - \frac{1}{5}$

$$= \frac{x}{4\sqrt{x^2 + 9}} - \frac{1}{5}$$

$$\frac{d^2T}{dx^2} = \frac{1}{4} \left[\frac{(x^2 + 9)^{1/2} \cdot 1 - x \cdot \frac{1}{2} \cdot 2x (x^2 + 9)^{-1/2}}{(x^2 + 9)} \right]$$

$$= \frac{\frac{1}{4} (x^2 + 9)^{-1/2} [x^2 + 9 - x^2]}{x^2 + 9}$$

$$= \frac{x^2 - x + 9}{4(x^2 + 9)\sqrt{x^2 + 9}}$$

At S.P. $\frac{dT}{dx} = 0$

$$\therefore 5x = 4\sqrt{x^2 + 9}$$

$$25x^2 = 16x^2 + 144$$

$$9x^2 = 144$$

$$x^2 = 16$$

$\Rightarrow$

$$\therefore \begin{cases} x = 4 & \text{or} & x = -4 \end{cases}$$

$$\begin{cases} T'' > 0 \end{cases}$$

$\Rightarrow$ min

NOT A SOLUTION

$$\therefore x \text{ is } 4 \text{ km,}$$

$$(b) (i) 1 - t + t^2 - \frac{t^3}{1+t}$$

$$= \frac{1+t - t - t^2 + t^2 + t^3 - t^3}{1+t}$$

$$= \frac{1}{1+t}$$

$$(ii) \int_0^x \frac{dt}{1+t} = \int_0^x dt - \int_0^x t dt + \int_0^x t^2 dt - \int_0^x \frac{t^3}{1+t} dt.$$

$$< \int_0^x dt - \int_0^x t dt + \int_0^x t^2 dt$$

$$= \left[t - \frac{1}{2}t^2 + \frac{1}{3}t^3 \right]_0^x$$

$$= x - \frac{1}{2}x^2 + \frac{1}{3}x^3.$$